Operations Research

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Exercise Sheet 1

Demo Exercises

Exercise D1. 1. *Toy production*

The Toys&Joys GmbH produces wind-up ducks and toy trains made of wood. One wind-up duck can be sold for €27. The material cost per duck is €10, the other variable costs are €14. The selling price for the toy train is €21, the material cost and other variable costs per toy train are €9 and €10, respectively.

Two operations are necessary in the production: Painting and assembling. Painting the individual parts of a duck takes 2 hours, their assembly 1 hour. For the train, painting and assembly each take 1 hour.

Any number of raw materials can be used each week. However, per week a maximum of 100 hours is available for painting, and a maximum of 80 hours is available for assembly. In addition, the demand for wind-up ducks is limited to 40 pieces per week.

Define a linear program whose solution maximizes the weekly profit under the given restrictions. Afterwards, solve this graphically.

Exercise D1. 2. *Fuel oil trade*

In his spare time, Steffen trades in fuel oil. For this purpose, he owns a tank with q liters capacity, in which he can temporarily store fuel oil for any length of time. For the coming n days, Steffen wants to design a buying and selling plan which maximizes his profit.

On each morning and evening of the n days, a tanker truck with a capacity of r liters arrives at Steffen's house. While Steffen can only buy (but not sell) fuel oil in the morning, he can only sell (but not buy) fuel oil in the evening. Since the tanker truck also targets other customers, it is already filled with g_i liters of fuel oil in the morning of the i-th day and with h_i liters of fuel oil in the evening of the i-th day, before it arrives at Steffen's house.

In addition, the fuel oil price is subject to daily fluctuation. Through extensive research, however, Steffen knows that on day i he can buy fuel oil at a price of b_i euros per liter and sell it for s_i euros per liter.

Steffen can temporarily store at most as much fuel oil as fits into his tank. Moreover, he cannot sell more fuel oil than he actually owns. Besides, a tanker truck can never take more fuel oil than fits into its tank and can sell at most as much fuel oil as it currently has in its tank.

Set up a linear program whose solution is a buying and selling plan which maximizes Steffen's profit. Assume that Steffen's tank is already filled with w liters of fuel oil at the beginning of the first day.

Exercise D1. 3. *Fitness*

Georg has signed up at the gym in order to be fully trained by the swimming season. His goal is to train the muscle groups $g = 1, \ldots, n$ and to reach a training level of T_g in each of them.

Georg can train up to 1000 minutes until the start of the swimming season. He has the option of either training alone or hiring a personal trainer. He has to pay 1 euro for every minute he is supervised by his personal trainer. The first 400 minutes that Georg trains without a trainer cost him nothing, and for each additional minute that he trains alone, he has to pay 0.10 euros.

In the studio m machines $M_1, \ldots, M_m$ are available for Georg to train with. If he trains alone on machine M_j, every minute of training contributes a_{jg} to training level T_g; every minute of training under the guidance of the personal trainer on M_j contributes p_{jg}.

Create a training plan for Georg so that he achieves his desired level of training in all muscle groups in the time available to him, while minimizing his costs.